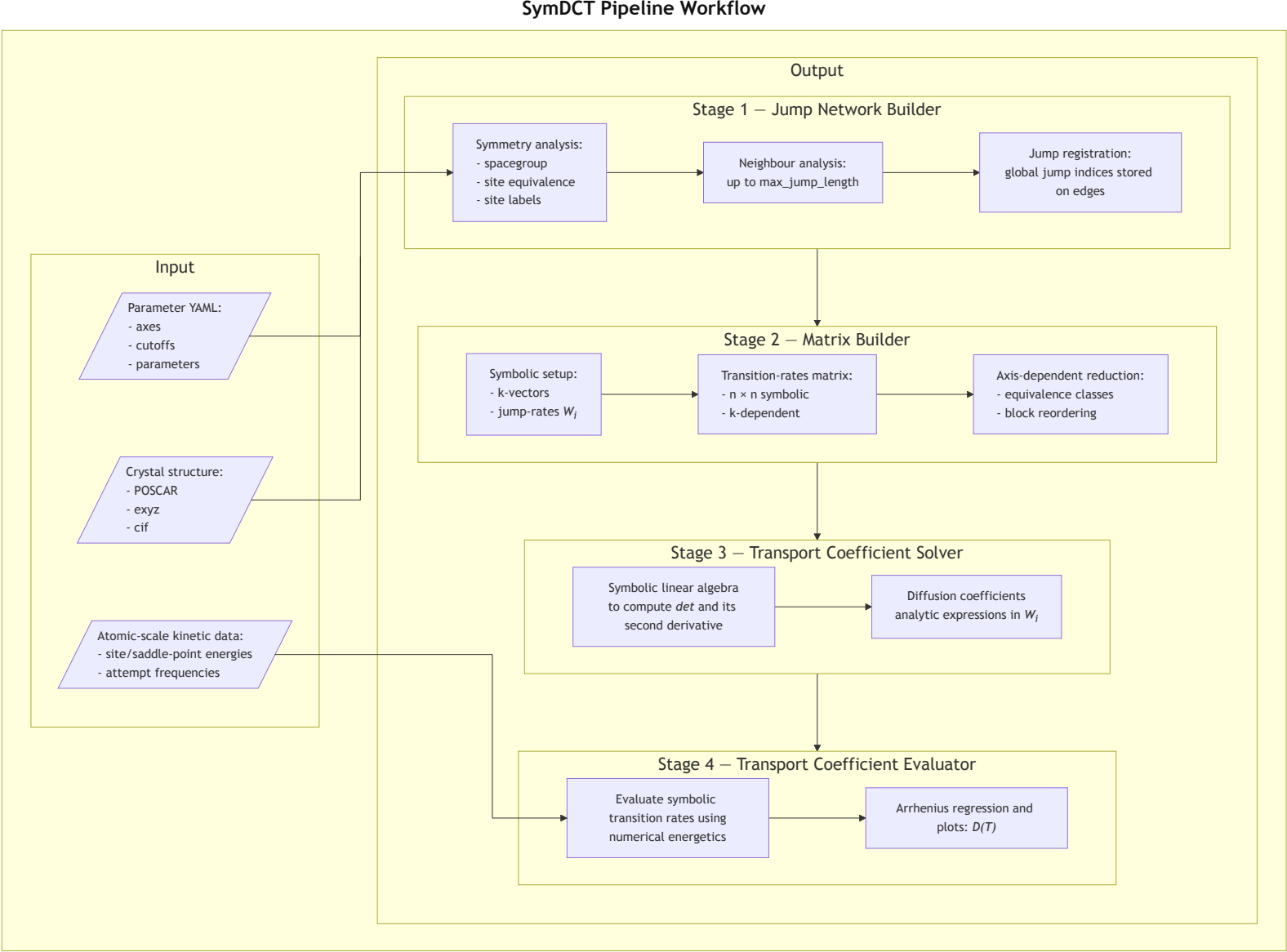

SymDCT Pipeline Workflow
Input
Parameter YAML:
- axes
- cutoffs
- parameters
Crystal structure:
- POSCAR
- exyz
- cif
Atomic-scale kinetic data:
- site/saddle-point energies
- attempt frequencies
Output
Stage 1 — Jump Network Builder
Symmetry analysis:
- spacegroup
- site equivalence
- site labels
Neighbour analysis:
up to max_jump_length
Jump registration:
global jump indices stored
on edges
Stage 2 — Matrix Builder
Symbolic setup:
- k-vectors
- jump-rates W_i
Transition-rates matrix:
- n × n symbolic
- k-dependent
Axis-dependent reduction:
- equivalence classes
- block reordering
Stage 3 — Transport Coefficient Solver
Symbolic linear algebra
to compute *det* and its
second derivative
Diffusion coefficients
analytic expressions in W_i
Stage 4 — Transport Coefficient Evaluator
Evaluate symbolic
transition rates using
numerical energetics
Arrhenius regression and
plots: $D(T)$